The Self-Sustaining Theory of Epistemology

A Unified Dynamic Theory of
Knowledge, Information, Epistemic Labor, Defense, Mobility, and Institutional Survival

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a unified Self-Sustaining Theory of Epistemology (SSTE), formulated as an open dynamic epistemic system with N epistemologists, where epistemologist i is associated with n_i workers. The theory addresses not merely the production of knowledge but the reproduction of the institutional capacity required to produce, validate, preserve, communicate, defend, criticize, revise, and reproduce knowledge.

The central distinction is between information and knowledge. Information is the broader set of propositions, observations, measurements, hypotheses, arguments, reports, computational outputs, and communications entering the epistemic system. Knowledge is information that has satisfied the relevant standards of validation. Propaganda is defined operationally as communicated information that has not attained validated knowledge status at the time of communication. This definition deliberately avoids equating propaganda with deliberate deception: conjectures, preliminary findings, criticisms, anomalies, and competing hypotheses may all initially be non-knowledge.

Because epistemic agents have finite processing capacity, non-validated information generates an epistemic burden. The theory therefore treats information production as an economic allocation problem in which the marginal epistemic value of information must be balanced against its marginal processing cost. Workers are endogenous epistemic inputs, and the paper derives their optimal allocation across epistemologists.

The SSTE additionally requires epistemic defense. Epistemic weapons are methodological instruments capable of testing, falsifying, challenging, correcting, or neutralizing invalid inference. Proof, counterexample, falsification, replication, source criticism, statistical testing, adversarial review, red-team analysis, dimensional analysis, and formal verification are examples. The Non-Immunity Principle requires that the same epistemic weapons used against external claims remain applicable to the SSTE itself.

Self-sustainability is distinguished from self-containment. Internal reproduction is necessary for institutional continuity, but external entry is necessary to maintain permeability to independent ideas, methods, evidence, and criticism. Conversely, epistemic exit is necessary to preserve meaningful institutional freedom and prevent epistemic captivity. Re-entry completes the mobility cycle.

The resulting system is an open, recursive, dynamically constrained epistemic economy. Its viability depends upon the joint satisfaction of knowledge, capacity, defense, and mobility conditions. Its long-run stability can be characterized through the Jacobian of the system's state transition function. The unified theory therefore connects epistemology with economics, accounting, finance, statistics, information theory, network science, computer science, artificial intelligence, psychology, political science, diplomacy, international relations, welfare economics, and institutional design.

The paper ends with "The End"

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1 Introduction

A conventional theory of epistemology asks questions such as:

$$\text{What is knowledge?} \tag{1}$$
$$\text{How is knowledge justified?} \tag{2}$$
$$\text{How can belief be distinguished from truth?} \tag{3}$$

A theory of a persistent epistemic institution must ask an additional question:

$$\boxed{\text{How does a system reproduce the capacity to produce knowledge?}} \tag{4}$$

This paper addresses that question.
Let there be

$$N(t) \in \mathbb{N} \tag{5}$$

epistemologists at time t.
Let

$$n_i(t) \in \mathbb{N}_0, \qquad i = 1, \ldots, N(t), \tag{6}$$

denote the number of workers associated with epistemologist i.
Total workers are

$$W(t) = \sum_{i=1}^{N(t)} n_i(t). \tag{7}$$

Total incumbent membership is

$$M(t) = N(t) + W(t). \tag{8}$$

The fundamental proposition of the SSTE is

$$\boxed{\text{The ultimate product of epistemology is the continuously reproduced capacity to produce knowledge.}} \tag{9}$$

This requires a system capable of:

$$\boxed{\text{production} \to \text{validation} \to \text{preservation} \to \text{communication} \to \text{defense} \to \text{correction} \to \text{reproduction.}} \tag{10}$$

But because no epistemic institution possesses perfect knowledge, this internal cycle must remain connected to an external epistemic environment.

Hence the complete architecture is

$$\boxed{\text{Entry} \longleftrightarrow \text{SSTE} \longleftrightarrow \text{Exit.}} \tag{11}$$

2 The Epistemic Ontology

2.1 Information

Let

$$\mathcal{I}(t) \tag{12}$$

denote the information entering or circulating through the SSTE during period t.

An information item may be represented by

$$x = (\text{claim}, \text{source}, \text{evidence}, \text{timestamp}, \text{status}). \tag{13}$$

Information may include:

$$\text{observations}, \tag{14}$$
$$\text{measurements}, \tag{15}$$
$$\text{hypotheses}, \tag{16}$$
$$\text{arguments}, \tag{17}$$
$$\text{computational outputs}, \tag{18}$$
$$\text{predictions}, \tag{19}$$
$$\text{criticisms}, \tag{20}$$
$$\text{reports}, \tag{21}$$
$$\text{anomalies}. \tag{22}$$

2.2 Knowledge

Let

$$\mathcal{K}(t) \tag{23}$$

denote validated epistemic information accumulated by time t.

The scalar knowledge stock is

$$K(t) \geq 0. \tag{24}$$

Newly validated knowledge during period t is

$$\Delta K(t). \tag{25}$$

The distinction is fundamental:

$$\boxed{K(t) \neq \Delta K(t).} \tag{26}$$

Knowledge accumulation follows

$$K(t+1) = K(t) + \Delta K(t) - \delta_K(t), \tag{27}$$

where $\delta_K(t)$ represents knowledge depreciation, obsolescence, or loss.

2.3 Propaganda

Define propaganda operationally as communicated information that has not attained validated knowledge status at the time of communication.

Thus

$$\boxed{x \in \mathcal{P}(t) \Longleftrightarrow x \in \mathcal{I}(t) \text{ and} x \notin \mathcal{K}(t)} \tag{28}$$

at the relevant time of communication.

This definition deliberately avoids the proposition

$$\text{propaganda} = \text{deliberate falsehood}. \tag{29}$$

A conjecture may be propaganda under the operational definition and subsequently become knowledge.

Similarly, a proposition may be false without being deliberately deceptive.

Therefore

$$\boxed{\text{Non-Knowledge} \neq \text{Falsehood} \neq \text{Deception.}} \tag{30}$$

3 The Epistemic Labor Structure

For epistemologist i, define the worker set

$$\mathcal{W}_i(t) = \left\{w_{i1}, \ldots, w_{in_i(t)}\right\}. \tag{31}$$

The full institutional population is

$$\mathcal{M}(t) = \mathcal{E}(t) \cup \bigcup_{i=1}^{N(t)} \mathcal{W}_i(t). \tag{32}$$

Epistemologists may perform:

$$G_i = \text{question and hypothesis generation,} \tag{33}$$
$$T_i = \text{testing,} \tag{34}$$
$$V_i = \text{validation,} \tag{35}$$
$$S_i = \text{preservation and transmission,} \tag{36}$$
$$R_i = \text{reproduction,} \tag{37}$$
$$D_i = \text{defense.} \tag{38}$$

Workers support these functions through informational, empirical, analytical, computational, administrative, communicative, and technical labor.

4 Information Production

Let gross information production associated with epistemologist i be

$$q_i(n_i) = \alpha_i n_i^{\beta_i}, \tag{39}$$

where

$$\alpha_i > 0, \qquad 0 < \beta_i < 1. \tag{40}$$

The first derivative is

$$q_i'(n_i) = \alpha_i \beta_i n_i^{\beta_i - 1} > 0, \tag{41}$$

while

$$q_i''(n_i) = \alpha_i \beta_i (\beta_i - 1) n_i^{\beta_i - 2} < 0. \tag{42}$$

Thus worker production exhibits diminishing marginal returns.

Aggregate gross information production is

$$Q(\mathbf{n}) = \sum_{i=1}^{N} \alpha_i n_i^{\beta_i}. \tag{43}$$

5 Knowledge Conversion

Let

$$\theta_i \in [0, 1] \tag{44}$$

denote the expected knowledge-conversion efficiency associated with epistemologist i.
Then

$$\Delta K(\mathbf{n}) = \sum_{i=1}^{N} \theta_i \alpha_i n_i^{\beta_i}. \tag{45}$$

The non-validated component is

$$P(\mathbf{n}) = \sum_{i=1}^{N} (1 - \theta_i) \alpha_i n_i^{\beta_i}. \tag{46}$$

Hence

$$\boxed{Q(\mathbf{n}) = \Delta K(\mathbf{n}) + P(\mathbf{n}).} \tag{47}$$

This equation formalizes the proposition that workers must be capable of producing both knowledge and non-knowledge.

6 Epistemic Processing Capacity

Let

$$C(t) > 0 \tag{48}$$

denote aggregate epistemic processing capacity.
It encompasses:

$$\text{attention,} \tag{49}$$
$$\text{research infrastructure,} \tag{50}$$
$$\text{computational resources,} \tag{51}$$
$$\text{validation labor,} \tag{52}$$
$$\text{institutional memory,} \tag{53}$$
$$\text{communication capacity,} \tag{54}$$
$$\text{defensive capacity,} \tag{55}$$
$$\text{training capacity.} \tag{56}$$

The information-flow constraint is

$$Q(t) \leq C(t). \tag{57}$$

Knowledge and capacity must not be conflated:

$$\boxed{K(t) \neq C(t).} \tag{58}$$

A system can possess extensive accumulated knowledge while simultaneously losing the capacity required to reproduce or verify it.

7 Epistemic Burden

Let the burden generated by non-validated information be

$$B(P) = bP^{\gamma}, \qquad b > 0, \quad \gamma > 1. \tag{59}$$

Then

$$B'(P) = b\gamma P^{\gamma-1} > 0 \tag{60}$$

and

$$B''(P) = b\gamma(\gamma - 1)P^{\gamma-2} > 0. \tag{61}$$

Thus the marginal burden of non-validated information increases with the information load.
This produces the central informational trade-off:

$$\boxed{\text{More Information} \to \text{More Potential Knowledge} + \text{More Processing Burden.}} \tag{62}$$

8 The Optimal Information Principle

Suppose net epistemic output is

$$\Phi(P) = F(P) - B(P), \tag{63}$$

where $F(P)$ is the expected knowledge value generated by information.
An interior optimum satisfies

$$\Phi'(P^*) = 0. \tag{64}$$

Therefore

$$\boxed{F'(P^*) = B'(P^*).} \tag{65}$$

The optimal information environment is consequently determined by equality between marginal epistemic benefit and marginal epistemic burden.
The theory does not therefore imply either

$$P = 0 \tag{66}$$

or

$$P \to \infty. \tag{67}$$

Instead, an interior optimum may exist:

$$0 < P^* < \infty. \tag{68}$$

9 Worker Allocation

Let the cost of employing worker j under epistemologist i be represented by the constant marginal cost $c_i > 0$.
Total worker cost is

$$C_W(\mathbf{n}) = \sum_{i=1}^{N} c_i n_i. \tag{69}$$

The static institutional objective is

$$\Pi_E(\mathbf{n}) = \Delta K(\mathbf{n}) - B(P(\mathbf{n})) - C_W(\mathbf{n}). \tag{70}$$

Thus

$$\boxed{\Pi_E = \text{validated knowledge} - \text{epistemic burden} - \text{worker cost.}} \tag{71}$$

The institution chooses

$$\mathbf{n}^* = \arg\max_{\mathbf{n}\geq 0} \Pi_E(\mathbf{n}). \tag{72}$$

10 Optimal Worker-Allocation Condition

For an interior solution,

$$\frac{\partial \Pi_E}{\partial n_i} = 0. \tag{73}$$

Since

$$\frac{\partial \Delta K}{\partial n_i} = \theta_i \alpha_i \beta_i n_i^{\beta_i - 1}, \tag{74}$$

and

$$\frac{\partial P}{\partial n_i} = (1-\theta_i)\alpha_i \beta_i n_i^{\beta_i - 1}, \tag{75}$$

we obtain

$$\boxed{\alpha_i \beta_i n_i^{\beta_i - 1} \left[\theta_i - (1-\theta_i)B'(P)\right] = c_i.} \tag{76}$$

Define effective epistemic productivity by

$$\psi_i(P) = \theta_i - (1-\theta_i)B'(P). \tag{77}$$

Then

$$\boxed{\alpha_i \beta_i \psi_i(P) n_i^{\beta_i - 1} = c_i.} \tag{78}$$

An interior solution requires

$$\psi_i(P) > 0. \tag{79}$$

Therefore

$$B'(P) < \frac{\theta_i}{1-\theta_i} \tag{80}$$

when $\theta_i < 1$.

11 Propaganda Threshold

The threshold $P_i^\dagger$ satisfies

$$\theta_i = (1-\theta_i)B'(P_i^\dagger). \tag{81}$$

For

$$B(P) = bP^\gamma, \tag{82}$$

this gives

$$\boxed{P_i^\dagger = \left[\frac{\theta_i}{(1-\theta_i)b\gamma}\right]^{\frac{1}{\gamma-1}}.} \tag{83}$$

Therefore

$$P < P_i^{\dagger} \tag{84}$$

is required for positive effective epistemic productivity.

The threshold rises with conversion efficiency:

$$\frac{\partial P_i^{\dagger}}{\partial \theta_i} > 0. \tag{85}$$

An epistemologist who converts a larger fraction of information into validated knowledge can therefore process a larger non-validated information environment before the marginal burden becomes prohibitive.

12 Epistemic Defense

The SSTE must defend both its agents and its epistemic procedures.

Let the epistemic arsenal of epistemologist i be

$$\mathcal{A}_i = a_{i1}, \dots, a_{im_i}. \tag{86}$$

Examples are shown in Table 1.

Table 1: Representative Epistemic Weapons

Method	**Primary Function**	**Failure Detected**
Proof	Establish logical consequence	Invalid deduction
Counterexample	Refute universal claims	Overgeneralization
Falsification	Seek empirical rejection	False empirical claims
Replication	Test reproducibility	Fragility
Source criticism	Verify provenance	Fabrication
Consistency analysis	Detect contradiction	Incoherence
Statistical testing	Quantify uncertainty	Inferential error
Adversarial review	Search for weaknesses	Confirmation bias
Red-team analysis	Stress-test systems	Structural vulnerability
Dimensional analysis	Check units	Physical inconsistency
Formal verification	Check computation	Algorithmic error

The term “weapon” is methodological rather than physical. Its object is invalid inference.

13 Epistemic Attack

Let

$$A(t) \geq 0 \tag{87}$$

denote epistemic attack intensity.

Potential attacks include:

$$\text{false claims,} \tag{88}$$
$$\text{fabricated evidence,} \tag{89}$$
$$\text{source contamination,} \tag{90}$$
$$\text{statistical manipulation,} \tag{91}$$
$$\text{logical inconsistency,} \tag{92}$$
$$\text{information flooding,} \tag{93}$$
$$\text{institutional capture,} \tag{94}$$
$$\text{suppression of criticism.} \tag{95}$$

The attack need not successfully persuade an agent. It may instead consume scarce validation capacity.

Thus

$$A(t)\uparrow \implies B(t)\uparrow \implies C_{\text{effective}}(t)\downarrow \tag{96}$$

may arise through an information-flooding mechanism.

14 Defensive Capacity

Let

$$d_i(n_i) = \delta_i n_i^{\rho_i}, \tag{97}$$

where

$$\delta_i > 0, \qquad 0 < \rho_i \leq 1. \tag{98}$$

Aggregate defensive capacity is

$$D_C = \sum_{i=1}^{N} \delta_i n_i^{\rho_i}. \tag{99}$$

The defensive margin is

$$M_D = D_C - A. \tag{100}$$

The basic defense constraint is

$$\boxed{D_C \geq A.} \tag{101}$$

15 The Non-Immunity Principle

The SSTE must be capable of applying its own epistemic weapons against itself.

Let

$$\mathcal{A}_{\text{external}} \tag{102}$$

denote the methods used to challenge external claims and

$$\mathcal{A}_{\text{internal}} \tag{103}$$

the methods permitted against the SSTE.

The Non-Immunity Principle requires

$$\boxed{\mathcal{A}_{\text{external}} \subseteq \mathcal{A}_{\text{internal}}.} \tag{104}$$

Hence

$$\begin{align}
E_i &\text{ may challenge } E_j, \tag{105}\\
w_{ij} &\text{ may challenge } E_i, \tag{106}\\
E_i &\text{ may challenge institutional procedures}, \tag{107}\\
w_{ij} &\text{ may challenge institutional procedures}, \tag{108}\\
\mathcal{M} &\text{ may challenge the SSTE itself.} \tag{109}
\end{align}$$

This gives the stronger principle

$$\boxed{\text{No epistemic authority possesses permanent immunity from legitimate epistemic criticism.}} \tag{110}$$

16 Epistemic Sovereignty and Epistemic Humility

The SSTE requires institutional authority to evaluate propositions.

This constitutes epistemic sovereignty.

However,

$$\boxed{\text{Epistemic Sovereignty} \neq \text{Epistemic Infallibility.}} \tag{111}$$

The institution may therefore maintain standards without declaring itself incapable of error.

The desired feedback structure is

$$\boxed{\text{Claim} \to \text{Challenge} \to \text{Evidence} \to \text{Evaluation} \to \text{Correction.}} \tag{112}$$

17 Internal Reproduction

Let

$$R_E(t) = \text{new internally reproduced epistemologists,} \tag{113}$$
$$R_W(t) = \text{new internally reproduced workers.} \tag{114}$$

Total internal reproduction is

$$R(t) = R_E(t) + R_W(t). \tag{115}$$

Internal reproduction requires:

$$\text{training,} \tag{116}$$
$$\text{mentorship,} \tag{117}$$
$$\text{knowledge transmission,} \tag{118}$$
$$\text{institutional memory,} \tag{119}$$
$$\text{resources,} \tag{120}$$
$$\text{career progression.} \tag{121}$$

Self-sustainability therefore requires more than maintaining a static membership list.

$$\boxed{\text{Self-Sustainability} = \text{Reproduction of Epistemic Capacity.}} \tag{122}$$

18 External Entry

Let

$$V_E(t) = \text{new or visiting epistemologists,} \tag{123}$$
$$V_W(t) = \text{new or visiting workers.} \tag{124}$$

Total external entry is

$$V(t) = V_E(t) + V_W(t). \tag{125}$$

Known entry mechanisms include:

Table 2: Epistemic Entry Mechanisms

Mechanism	**Purpose**
Open application	External candidacy
Visiting appointment	Temporary participation
Fellowship	Structured integration
Peer review	Contribution-based admission
Independent replication	Demonstrated competence
Seminar participation	External criticism and exchange
External challenge	Direct testing of claims
Examination	Competence assessment
Probationary membership	Controlled integration
Interdisciplinary exchange	Methodological diversification
Secondment	Temporary institutional transfer

The central admission principle is

$$\boxed{\text{Epistemic Admission} \neq \text{Ideological Conformity.}} \tag{126}$$

Competence, evidence, reasoning, and relevant contribution may legitimately be evaluated. Mere agreement with incumbent conclusions should not constitute the sole admission criterion.

19 Anti-Echo-Chamber Condition

Let

$$H_t \tag{127}$$

denote the distribution of hypotheses, methodologies, and epistemic positions represented within the SSTE.

Let

$$\mathcal{D}(H_t) \tag{128}$$

denote an epistemic-diversity functional.

An institution is structurally exposed to echo-chamber formation if its reproduction mechanism systematically contracts diversity:

$$\mathcal{D}(H_{t+1}) < \mathcal{D}(H_t). \tag{129}$$

Repeated contraction can produce

$$\mathcal{D}(H_t) \to 0. \tag{130}$$

External entry introduces an external distribution G_t:

$$H_{t+1} = (1 - \omega_t)H_t + \omega_t G_t, \qquad 0 < \omega_t < 1. \tag{131}$$

This does not guarantee diversity. It creates the possibility of independent variation.

The anti-echo-chamber condition is therefore not

$$V > 0 \tag{132}$$

alone.

Rather,

$$\boxed{\text{Entry must provide a credible possibility of epistemically independent variation.}} \tag{133}$$

20 Epistemic Exit

Let

$$X_E(t) = \text{epistemologists exiting,} \tag{134}$$
$$X_W(t) = \text{workers exiting.} \tag{135}$$

Total exit is

$$X(t) = X_E(t) + X_W(t). \tag{136}$$

Legitimate exit mechanisms include:

$$\text{voluntary resignation,} \tag{137}$$
$$\text{completion of a visiting appointment,} \tag{138}$$
$$\text{career transition,} \tag{139}$$
$$\text{independent research,} \tag{140}$$
$$\text{retirement,} \tag{141}$$
$$\text{temporary withdrawal,} \tag{142}$$
$$\text{transparent institutional separation.} \tag{143}$$

Exit must not automatically be interpreted as epistemic failure.

$$\boxed{\text{Institutional Exit} \neq \text{Epistemic Failure.}} \tag{144}$$

Nor does exit imply disappearance from the wider epistemic network.

$$\boxed{\text{Institutional Exit} \neq \text{Epistemic Disappearance.}} \tag{145}$$

21 Voice and Exit

Let V_a denote the value of remaining within the institution and X_a the value of leaving.
The agent should have meaningful access to both mechanisms.
Thus

$$\boxed{\text{Voice} + \text{Exit} = \text{Epistemic Freedom.}} \tag{146}$$

If criticism is possible but exit is impossible, institutional authority may become coercive.
If exit is possible but criticism is suppressed, internal correction becomes weaker.
A resilient SSTE requires both.

22 Re-entry

Let

$$V_E^R(t) = \text{returning epistemologists,} \tag{147}$$
$$V_W^R(t) = \text{returning workers.} \tag{148}$$

Re-entry produces the complete mobility cycle:

$$\boxed{\text{Entry} \to \text{Participation} \to \text{Exit} \to \text{Independent Activity} \to \text{Re-entry.}} \tag{149}$$

The SSTE is consequently connected to a broader epistemic ecosystem rather than functioning as a closed population.

23 Membership Dynamics

Epistemologist membership evolves according to

$$N(t+1) = N(t) + R_E(t) + V_E(t) + V_E^R(t) - X_E(t). \quad (150)$$

Worker membership evolves according to

$$W(t+1) = W(t) + R_W(t) + V_W(t) + V_W^R(t) - X_W(t). \quad (151)$$

Therefore total membership satisfies

$$\boxed{M(t+1) = M(t) + R(t) + V(t) + V^R(t) - X(t).} \quad (152)$$

Population growth requires

$$R(t) + V(t) + V^R(t) > X(t). \quad (153)$$

Population contraction occurs when

$$R(t) + V(t) + V^R(t) < X(t). \quad (154)$$

A stationary membership equilibrium satisfies

$$\boxed{R^* + V^* + V^{R*} = X^*.} \quad (155)$$

24 Epistemic Capacity Dynamics

Let

$$O(t) \quad (156)$$

denote useful epistemic capacity contributed by external entrants and

$$C_V(t) \quad (157)$$

the corresponding integration and validation cost.
Let

$$D(t) \quad (158)$$

denote epistemic depreciation.
Then

$$\boxed{C(t+1) = C(t) + R_C(t) + O(t) - D(t) - B(t) - C_V(t).} \quad (159)$$

The capacity margin is

$$M_C(t) = R_C(t) + O(t) - D(t) - B(t) - C_V(t). \quad (160)$$

Non-declining capacity requires

$$\boxed{M_C(t) \geq 0.} \quad (161)$$

Strictly positive capacity growth requires

$$M_C(t) > 0. \quad (162)$$

25 Self-Sustainability Theorem

Theorem. Suppose the SSTE has capacity dynamics

$$C(t+1) = C(t) + R_C(t) + O(t) - D(t) - B(t) - C_V(t). \tag{163}$$

If, for all sufficiently large t,

$$R_C(t) + O(t) \geq D(t) + B(t) + C_V(t), \tag{164}$$

then

$$C(t+1) \geq C(t). \tag{165}$$

Proof.
Subtracting $C(t)$ from the capacity equation gives

$$C(t+1) - C(t) = R_C(t) + O(t) - D(t) - B(t) - C_V(t). \tag{166}$$

The assumed inequality implies

$$C(t+1) - C(t) \geq 0. \tag{167}$$

Therefore

$$C(t+1) \geq C(t). \tag{168}$$

Thus capacity is non-decreasing. Strict inequality in the assumed condition produces positive capacity growth. □

26 Knowledge Dynamics

Knowledge has a separate law of motion:

$$K(t+1) = K(t) + \Delta K(t) - \delta_K(t). \tag{169}$$

The knowledge margin is

$$M_K(t) = \Delta K(t) - \delta_K(t). \tag{170}$$

Hence

$$M_K(t) > 0 \Rightarrow K(t+1) > K(t), \tag{171}$$
$$M_K(t) = 0 \Rightarrow K(t+1) = K(t), \tag{172}$$
$$M_K(t) < 0 \Rightarrow K(t+1) < K(t). \tag{173}$$

A stationary knowledge equilibrium satisfies

$$\boxed{\Delta K^* = \delta_K^*.} \tag{174}$$

A growing knowledge regime satisfies

$$\Delta K^* > \delta_K^*. \tag{175}$$

27 Defense Dynamics

Let defensive capacity evolve according to

$$D_C(t+1) = D_C(t) + I_D(t) + R_D(t) - \delta_D(t), \tag{176}$$

where:

$$I_D(t) = \text{defensive investment}, \tag{177}$$
$$R_D(t) = \text{defensive capacity generated through reproduction}, \tag{178}$$
$$\delta_D(t) = \text{defensive depreciation}. \tag{179}$$

The defensive margin is

$$M_D(t) = D_C(t) - A(t). \tag{180}$$

A viable defense requires

$$\boxed{M_D(t) \geq 0.} \tag{181}$$

Robust defense requires a positive margin:

$$M_D(t) > 0. \tag{182}$$

28 The Four Principal SSTE Margins

The unified theory contains four fundamental margins.
Knowledge:

$$M_K = \Delta K - \delta_K. \tag{183}$$

Capacity:

$$M_C = R_C + O - D - B - C_V. \tag{184}$$

Defense:

$$M_D = D_C - A. \tag{185}$$

Membership:

$$M_M = R + V + V^R - X. \tag{186}$$

Collectively,

$$\boxed{\mathbf{M} = \begin{bmatrix} M_K \ M_C \ M_D \ M_M \end{bmatrix}.} \tag{187}$$

These margins distinguish four different notions of institutional health.

29 Viability

Define the viability region

$$\mathcal{V} = \{\mathbf{S} : K \geq 0, ; C > 0, ; M > 0, ; M_C \geq 0, ; M_D \geq 0\}. \tag{188}$$

An SSTE is viable over a horizon $\mathcal{T}$ if

$$\mathbf{S}_t \in \mathcal{V} \tag{189}$$

for every

$$t \in \mathcal{T}. \tag{190}$$

This is stronger than requiring a single favorable period.

30 The Complete SSTE State Vector

Define

$$\mathbf{S}(t) = \left[K(t)\ C(t)\ N(t)\ W(t)\ P(t)\ B(t)\ D_C(t)\ A(t)\ V(t)\ X(t)\ D_{\mathcal{E}}(t)\right]. \tag{191}$$

The unified dynamic system can be represented as

$$\boxed{\mathbf{S}(t+1) = \mathcal{F}\left(\mathbf{S}(t), \mathbf{n}(t), \mathbf{u}(t), \boldsymbol{\varepsilon}(t)\right),} \tag{192}$$

where $\mathbf{u}(t)$ contains institutional decisions and $\boldsymbol{\varepsilon}(t)$ contains exogenous shocks.

31 Equilibrium

An SSTE equilibrium is a configuration

$$(\mathbf{S}^*, \mathbf{n}^*, \mathbf{u}^*) \tag{193}$$

such that:

1. worker allocations satisfy the relevant optimality conditions;
2. knowledge accumulation is internally consistent;
3. capacity accumulation is internally consistent;
4. defensive capacity satisfies the defense constraint;
5. entry decisions are consistent with external opportunity costs;
6. exit decisions are consistent with external opportunity values;
7. institutional accounting identities hold;
8. expectations are dynamically consistent.

A stationary equilibrium satisfies

$$\boxed{\mathbf{S}^* = \mathcal{F}\left(\mathbf{S}^*, \mathbf{n}^*, \mathbf{u}^*, \boldsymbol{\varepsilon}^*\right).} \tag{194}$$

32 Entry Equilibrium

Let G_E denote the expected institutional gain from admitting an external epistemologist and C_E^{entry} the expected integration cost.

Entry is optimal when

$$G_E > C_E^{\text{entry}}. \tag{195}$$

An interior entry equilibrium satisfies

$$\boxed{G_E = C_E^{\text{entry}}.} \tag{196}$$

The same condition can be formulated for workers.

Thus openness need not imply indiscriminate admission.

33 Exit Equilibrium

Let G_R denote the expected value of remaining inside and G_X the expected value of leaving.

An agent exits when

$$G_X > G_R. \tag{197}$$

An interior mobility equilibrium satisfies

$$\boxed{G_X = G_R.} \tag{198}$$

The SSTE therefore treats mobility as an endogenous economic and epistemic decision.

34 Accounting Representation

Let epistemic assets be

$$A_E(t) \tag{199}$$

and epistemic liabilities and costs be

$$L_E(t). \tag{200}$$

Define net epistemic capital as

$$E_N(t) = A_E(t) - L_E(t). \tag{201}$$

Epistemic assets include:

$$\text{validated knowledge,} \tag{202}$$
$$\text{trained epistemologists,} \tag{203}$$
$$\text{trained workers,} \tag{204}$$
$$\text{methods,} \tag{205}$$
$$\text{datasets,} \tag{206}$$
$$\text{institutional memory,} \tag{207}$$
$$\text{research infrastructure,} \tag{208}$$
$$\text{defensive capability.} \tag{209}$$

Epistemic liabilities include:

$$\text{knowledge depreciation,} \tag{210}$$
$$\text{personnel loss,} \tag{211}$$
$$\text{maintenance requirements,} \tag{212}$$
$$\text{data degradation,} \tag{213}$$
$$\text{methodological obsolescence,} \tag{214}$$
$$\text{propaganda burden,} \tag{215}$$
$$\text{integration costs.} \tag{216}$$

Thus

$$\boxed{\Delta E_N = \Delta A_E - \Delta L_E.} \tag{217}$$

This accounting representation is analytical rather than a claim that every epistemic object has a unique monetary price.

35 Finance and Intertemporal Investment

Let

$$I_C(t) \tag{218}$$

denote investment in epistemic capacity.
Then

$$C(t+1) = C(t) + I_C(t) + R_C(t) + O(t) - D(t) - B(t) - C_V(t). \tag{219}$$

With discount factor

$$0 < \delta < 1, \tag{220}$$

an institutional planner may solve

$$\max \sum_{t=0}^{\infty} \delta^t U\left(K(t), C(t), D_C(t), D_{\mathcal{E}}(t)\right). \tag{221}$$

The SSTE therefore possesses an intertemporal capital-allocation problem analogous to other durable institutions.

36 Welfare Economics

Let social welfare be

$$U_S = U_S\left(K, C, D_C, D_{\mathcal{E}}, P, V, X\right). \tag{222}$$

Potentially,

$$\frac{\partial U_S}{\partial K} > 0 \tag{223}$$

while

$$\frac{\partial U_S}{\partial B} < 0. \tag{224}$$

Information therefore creates both positive and negative externalities.
An individual's informational output may benefit the entire epistemic community while imposing validation costs upon others.
This produces a public-good problem:

$$\boxed{\text{Private Information Production} \to \text{Social Knowledge Benefit} + \text{Social Validation Cost.}} \tag{225}$$

37 Psychology and Cognitive Load

Let cognitive load on epistemologist i be

$$L_i = L_i^K + L_i^P + L_i^O, \tag{226}$$

where the terms denote knowledge-processing, propaganda-processing, and other cognitive loads.
Let error probability be

$$\Pr(\text{Error}_i) = g(L_i). \tag{227}$$

Over an appropriate range,

$$g'(L_i) > 0. \tag{228}$$

Thus excessive information can reduce epistemic accuracy even when the information itself is potentially valuable.
This provides a micro-foundation for the convex burden function.

38 Statistics and Bayesian Updating

Let H denote a hypothesis and E evidence.

Bayesian updating gives

$$\Pr(H \mid E) = \frac{\Pr(E \mid H)\Pr(H)}{\Pr(E)}. \tag{229}$$

For competing hypotheses,

$$\Pr(H_k \mid E) = \frac{\Pr(E \mid H_k)\Pr(H_k)}{\sum_{\ell=1}^{m} \Pr(E \mid H_\ell)\Pr(H_\ell)}. \tag{230}$$

The SSTE does not require Bayesian inference in every domain. Rather, Bayesian inference constitutes one epistemic weapon among several.

The priors, likelihoods, measurements, and model assumptions themselves remain challengeable.

39 Information Theory

Let informational entropy be

$$H(X) = -\sum_x p(x)\log p(x). \tag{231}$$

High entropy can represent informational diversity but does not automatically imply knowledge.

Thus

$$\boxed{\text{Information Entropy} \neq \text{Epistemic Value.}} \tag{232}$$

The relevant institutional problem is not maximizing information entropy without limit.

It is maximizing useful information subject to processing constraints.

40 Network Science

Represent the epistemic system as

$$G(t) = (V(t), A(t)), \tag{233}$$

where vertices are epistemic agents and edges represent communication, mentorship, collaboration, criticism, evidence transfer, or validation.

Redundant paths provide resilience.

A single-point failure can be represented by

$$\deg(v) = 1 \tag{234}$$

for a critical vertex v.

A highly centralized network can therefore exhibit

$$\text{centralization} \uparrow \implies \text{single-point vulnerability} \uparrow. \tag{235}$$

But excessive decentralization can increase coordination costs.

The SSTE therefore faces a structural trade-off:

$$H^* = \arg\max_H \{\text{coordination benefit} - \text{concentration cost}\}. \tag{236}$$

41 Computer Science and Provenance

Every significant epistemic object should, where feasible, carry provenance:

$$x = (\text{claim}, \text{source}, \text{evidence}, \text{time}, \text{status}). \tag{237}$$

Its status may be

$$s(x) \in \{\text{unverified}, \text{supported}, \text{validated}, \text{falsified}, \text{disputed}, \text{obsolete}\}. \tag{238}$$

The provenance graph is

$$x_0 \to x_1 \to x_2 \to \cdots \to x_T. \tag{239}$$

Auditable provenance reduces the cost of epistemic reconstruction.

42 Artificial Intelligence as Epistemic Labor

AI systems can function as workers.

Let AI worker j generate

$$q_j = k_j + p_j. \tag{240}$$

Increasing AI output implies

$$\frac{\partial Q}{\partial N_{\text{AI}}} > 0, \tag{241}$$

but does not imply

$$\frac{\partial K}{\partial N_{\text{AI}}} > 0. \tag{242}$$

Therefore

$$\boxed{\text{AI Output} \neq \text{Knowledge.}} \tag{243}$$

AI can nevertheless increase epistemic capacity through:

$$\text{hypothesis generation,} \tag{244}$$
$$\text{counterexample generation,} \tag{245}$$
$$\text{simulation,} \tag{246}$$
$$\text{data analysis,} \tag{247}$$
$$\text{anomaly detection,} \tag{248}$$
$$\text{literature synthesis,} \tag{249}$$
$$\text{adversarial review.} \tag{250}$$

All AI-generated information remains subject to validation.

43 Political Science, Diplomacy, and Geopolitics

Epistemic institutions operate within strategic environments.

Let state s possess information

$$\mathcal{I}_s = \mathcal{K}_s + \mathcal{P}_s + \mathcal{N}_s, \tag{251}$$

where $\mathcal{N}_s$ denotes irrelevant or noisy information.

Strategic actors may possess incentives to manipulate information.

The SSTE must therefore distinguish:

$$\text{information,} \tag{252}$$
$$\text{signaling,} \tag{253}$$
$$\text{deception,} \tag{254}$$
$$\text{credible commitment,} \tag{255}$$
$$\text{uncertainty,} \tag{256}$$
$$\text{verification.} \tag{257}$$

Diplomatic epistemology becomes an optimization problem:

$$\max\{\text{useful information extracted} - \text{verification cost}\}. \tag{258}$$

44 Warfare Economics and Epistemic Resilience

Modern conflict may target information and decision systems without directly attacking physical infrastructure.

An epistemic attack may seek to:

$$\text{degrade trust,} \tag{259}$$
$$\text{flood validation channels,} \tag{260}$$
$$\text{distort measurements,} \tag{261}$$
$$\text{corrupt provenance,} \tag{262}$$
$$\text{induce institutional disagreement.} \tag{263}$$

The relevant response is epistemic resilience:

$$\boxed{\text{Attack} \to \text{Detection} \to \text{Isolation} \to \text{Verification} \to \text{Correction} \to \text{Recovery.}} \tag{264}$$

45 The SSTE Architecture

The complete system is represented in Figure 1.

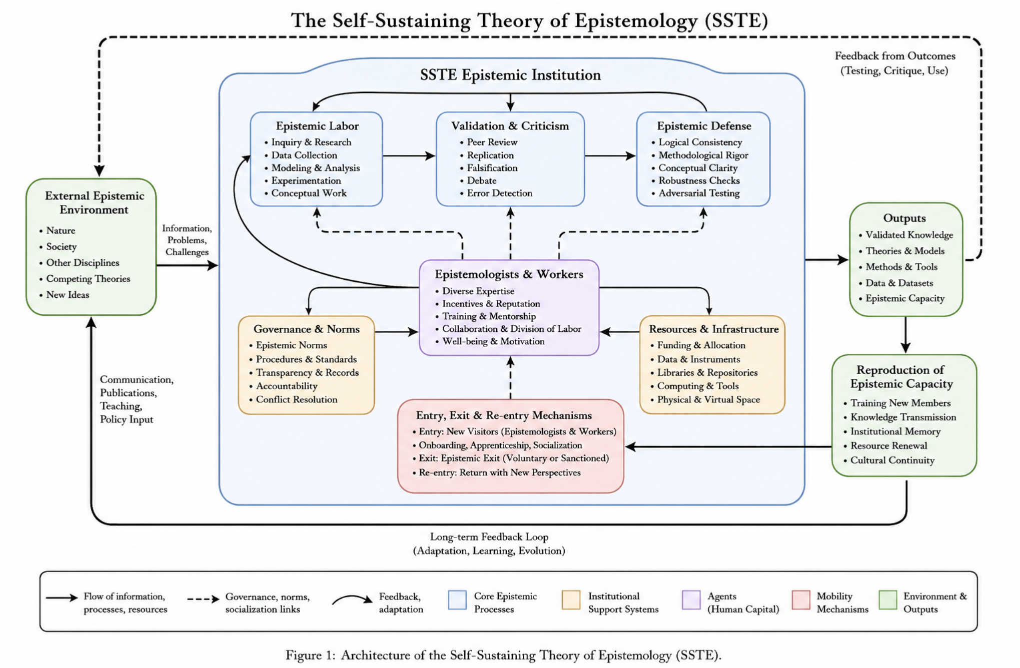

Figure 1: Unified architecture of the Self-Sustaining Theory of Epistemology.

46 The Epistemic Reproduction Cycle

The internal cycle is

$$\boxed{\text{Information} \to \text{Validation} \to \text{Knowledge} \to \text{Defense} \to \text{Correction} \to \text{Reproduction.}} \tag{265}$$

The external cycle is

$$\boxed{\text{Entry} \to \text{Novelty} \to \text{Testing} \to \text{Exit} \to \text{Independent Criticism} \to \text{Re-entry.}} \tag{266}$$

The two cycles jointly define the SSTE.

47 Local Stability

Suppose

$$\mathbf{S}_{t+1} = \mathcal{F}(\mathbf{S}_t). \tag{267}$$

Let

$$\mathbf{S}^* \tag{268}$$

be an equilibrium.
Define

$$\mathbf{z}_t = \mathbf{S}_t - \mathbf{S}^*. \tag{269}$$

A first-order approximation is

$$\mathbf{z}_{t+1} = J^* \mathbf{z}_t + O\left(|\mathbf{z}_t|^2\right), \tag{270}$$

where

$$J^* = \left.\frac{\partial \mathcal{F}}{\partial \mathbf{S}}\right|_{\mathbf{S}=\mathbf{S}^*}. \tag{271}$$

Local Stability Theorem.
If

$$\rho(J^*) < 1, \tag{272}$$

where ρ denotes the spectral radius, then the stationary equilibrium is locally asymptotically stable.
Proof.
The linearized system satisfies

$$\mathbf{z}_t = (J^*)^t \mathbf{z}_0. \tag{273}$$

If

$$\rho(J^*) < 1, \tag{274}$$

then

$$(J^*)^t \to 0 \tag{275}$$

as $t \to \infty$.
Therefore

$$\mathbf{z}_t \to 0, \tag{276}$$

so the trajectory converges locally to the equilibrium. □

48 Epistemic Resilience

Define the resilience vector

$$\mathbf{R} = \begin{bmatrix} M_K \\ M_C \\ M_D \\ D_{\mathcal{E}} \end{bmatrix}. \tag{277}$$

A locally resilient equilibrium requires:

$$M_C^* > 0, \tag{278}$$
$$M_D^* > 0, \tag{279}$$
$$\rho(J^*) < 1. \tag{280}$$

The strict margins provide local feasibility while the spectral condition provides dynamic recovery. Hence

$$\boxed{\text{Epistemic Resilience} = \text{Structural Margin} + \text{Dynamic Stability.}} \tag{281}$$

49 The Non-Sealing Principle

A self-sustaining epistemic system must not become self-sealing.

A self-sealing system systematically prevents one or more of:

$$\text{external entry,} \tag{282}$$
$$\text{internal criticism,} \tag{283}$$
$$\text{epistemic exit,} \tag{284}$$
$$\text{independent replication,} \tag{285}$$
$$\text{external verification.} \tag{286}$$

Therefore

$$\boxed{\text{Self-Sustaining} \neq \text{Self-Sealing.}} \tag{287}$$

Indeed, self-sealing may undermine self-sustainability because it eliminates mechanisms of error detection.

50 The Fundamental Openness Principle

The SSTE requires both internal reproduction and external permeability.

Thus

$$\boxed{\text{Endogenous Reproduction} + \text{Exogenous Permeability} = \text{Open Self-Sustainability.}} \tag{288}$$

Internal reproduction supplies continuity.
Entry supplies potential novelty.
Exit supplies independence and institutional feedback.
Re-entry supplies circulation.

51 The Unified SSTE Conditions

The SSTE requires six principal institutional conditions.

Condition I: Knowledge Production

$$\Delta K > 0 \tag{289}$$

over appropriate production intervals.

Condition II: Capacity Reproduction

$$R_C + O \geq D + B + C_V. \tag{290}$$

Condition III: Epistemic Defense

$$D_C \geq A. \tag{291}$$

Condition IV: Epistemic Non-Immunity

$$\mathcal{A}_{\text{external}} \subseteq \mathcal{A}_{\text{internal}}. \tag{292}$$

Condition V: Epistemic Entry

The institution must maintain a credible mechanism for external participation:

$$V > 0 \tag{293}$$

over relevant horizons.

Condition VI: Epistemic Exit

The institution must maintain a credible mechanism for departure:

$$X > 0 \tag{294}$$

over relevant horizons.
Together,

$$\boxed{\text{Production} \land \text{Reproduction} \land \text{Defense} \quad \land \text{Non-Immunity} \land \text{Entry} \land \text{Exit.}} \tag{295}$$

52 Central Propositions

The unified theory yields the following propositions.

52.1 Proposition 1: Knowledge Is a Stock

$$K(t) = K(t-1) + \Delta K(t) - \delta_K(t). \tag{296}$$

Knowledge must therefore be distinguished from knowledge production.

52.2 Proposition 2: Information Is an Input and Output

$$Q = \Delta K + P. \tag{297}$$

Information is therefore broader than knowledge.

52.3 Proposition 3: Non-Validated Information Has Both Value and Cost

$$P > 0 \tag{298}$$

may increase the probability of future discovery while simultaneously increasing

$$B(P). \tag{299}$$

52.4 Proposition 4: Worker Allocation Is Endogenous

The optimal allocation satisfies

$$\alpha_i \beta_i n_i^{\beta_i - 1} \left[\theta_i - (1-\theta_i) B'(P)\right] = c_i \tag{300}$$

when the solution is interior and unconstrained by additional institutional constraints.

52.5 Proposition 5: Defense Is Methodological

$$D_C \geq A \tag{301}$$

is a condition on epistemic resilience, not a mandate for physical violence.

52.6 Proposition 6: No Epistemic Authority Is Infallible

$$\mathcal{A}_{\text{external}} \subseteq \mathcal{A}_{\text{internal}}. \tag{302}$$

52.7 Proposition 7: Self-Sustainability Does Not Require Closure

$$\text{Self-Sustaining} \neq \text{Self-Contained}. \tag{303}$$

52.8 Proposition 8: Entry Does Not Guarantee Diversity

$$V > 0 \tag{304}$$

creates an opportunity for novelty but does not mechanically guarantee

$$\mathcal{D}(H_{t+1}) > \mathcal{D}(H_t). \tag{305}$$

52.9 Proposition 9: Exit Is Epistemically Productive

Exit may provide:

$$\begin{aligned} &\text{independent criticism,} && (306)\\ &\text{institutional information,} && (307)\\ &\text{external comparison,} && (308)\\ &\text{methodological diversification.} && (309) \end{aligned}$$

52.10 Proposition 10: Resilience Requires Both Margin and Stability

$$M_C > 0, \qquad M_D > 0, \qquad \rho(J^*) < 1 \tag{310}$$

jointly provide a local sufficient structure for resilient equilibrium.

53 A General SSTE Optimization Problem

The complete institutional planner's problem can be written as

$$\max_{\{\mathbf{n}_t, V_t, X_t, I_{C,t}, I_{D,t}\}} \sum_{t=0}^{\infty} \delta^t U\left(K_t, C_t, D_{C,t}, \mathcal{D}(H_t)\right) \tag{311}$$

subject to

$$
\begin{aligned}
K_{t+1} &= K_t + \Delta K_t - \delta_K(t), && (312)\\
C_{t+1} &= C_t + R_{C,t} + O_t - D_t - B_t - C_{V,t}, && (313)\\
N_{t+1} &= N_t + R_{E,t} + V_{E,t} + V_{E,t}^R - X_{E,t}, && (314)\\
W_{t+1} &= W_t + R_{W,t} + V_{W,t} + V_{W,t}^R - X_{W,t}, && (315)\\
Q_t &\leq C_t, && (316)\\
D_{C,t} &\geq A_t, && (317)\\
n_{i,t} &\geq 0. && (318)
\end{aligned}
$$

This establishes the SSTE as a dynamic optimization problem rather than merely a static classification of epistemic institutions.

54 Algorithmic SSTE Protocol

A generic epistemic protocol is:

$$
\begin{array}{ll}
1. & \text{Receive information item } x.\\
2. & \text{Record provenance.}\\
3. & \text{Identify epistemic domain.}\\
4. & \text{Identify applicable validation standard.}\\
5. & \text{Search for supporting evidence.}\\
6. & \text{Search for counterevidence.}\\
7. & \text{Attempt falsification.}\\
8. & \text{Check logical consistency.}\\
9. & \text{Check statistical validity.}\\
10. & \text{Replicate where feasible.}\\
11. & \text{Conduct adversarial review.}\\
12. & \text{Assign provisional epistemic status.}\\
13. & \text{Record uncertainty.}\\
14. & \text{Permit subsequent challenge.}\\
15. & \text{Update institutional knowledge and methods.}
\end{array} \tag{319}
$$

The crucial final step is institutional learning.

$$
\boxed{\text{The procedure must be capable of changing the procedure.}} \tag{320}
$$

55 Interdisciplinary Structure

Table 3: Interdisciplinary Interpretation of the SSTE

Field	**SSTE Contribution**
Mathematics	Formal models and equilibrium
Logic	Valid inference and contradiction detection
Economics	Production, incentives, scarcity, equilibrium
Finance	Investment and intertemporal allocation
Accounting	Stocks, flows, depreciation, capital
Physics	Dynamics, conservation, dissipation analogies
Chemistry	Transformation processes
Biology	Reproduction and adaptation
Psychology	Attention and cognitive load
Psychiatry	Institutional safeguards against cognitive vulnerability
Philosophy	Justification and skepticism
Religion	Plural epistemic traditions
Political science	Authority, voice, exit, institutions
Geopolitics	Strategic information environments
Diplomacy	Signaling and verification
International relations	Information asymmetry
Welfare economics	Externalities and social value
Warfare economics	Information resilience
Statistics	Uncertainty and inference
Computer science	Provenance and formal verification
Data science	Data validation
AI/ML	Automated epistemic labor
Network science	Connectivity and redundancy
Information theory	Information capacity
Game theory	Strategic communication
Control theory	Feedback and stability
Operations research	Resource allocation

56 The Fundamental SSTE Identity

The unified theory can be compressed into the following identity:

$$\boxed{\begin{aligned}\text{SSTE Survival} \iff & \text{Knowledge Production} + \text{Capacity Reproduction}\\ & + \text{Epistemic Defense} + \text{Institutional Openness}\\ & - \text{Depreciation} - \text{Epistemic Burden} - \text{Uncorrected Error} > 0.\end{aligned}} \tag{321}$$

The corresponding institutional architecture is

$$\boxed{\begin{aligned}&\text{Entry} \to \text{Epistemic Labor} \to \text{Information}\\ &\to \text{Validation} \to \text{Knowledge}\\ &\to \text{Defense} \to \text{Correction}\\ &\to \text{Reproduction} \to \text{Renewed Capacity}\\ &\to \text{Entry/Exit/Re-entry}.\end{aligned}} \tag{322}$$

57 Conclusion

The Self-Sustaining Theory of Epistemology is a theory of an epistemic institution that must continuously reproduce the capacity to discover, test, validate, preserve, communicate, defend, challenge, revise, and transmit knowledge.

Its starting point is deliberately simple:

$$N \text{ epistemologists}, \qquad n_i \text{ workers per epistemologist.} \tag{323}$$

From this structure follows an entire dynamic epistemic economy.
Workers produce information.
Some information becomes knowledge.
Some remains non-validated.
Non-validated information is necessary because discovery often precedes validation, but excessive non-validated information creates a processing burden.
Consequently, the optimal information environment is determined by the marginal trade-off

$$\boxed{\text{Marginal Epistemic Benefit} = \text{Marginal Epistemic Burden.}} \tag{324}$$

Worker allocation is endogenous and governed by marginal epistemic productivity net of informational congestion.
The SSTE must also defend itself.
Its epistemic weapons are the methods by which invalid inference is attacked: proof, falsification, counterexample, replication, source criticism, statistical testing, adversarial review, red-team analysis, dimensional analysis, and formal verification.
Yet the most important weapon is the weapon turned inward.

$$\boxed{\text{The SSTE must be capable of proving itself wrong.}} \tag{325}$$

This is the operational meaning of the Non-Immunity Principle.
Self-sustainability must not become self-sealing.
Internal reproduction provides continuity, but external entry provides potential novelty. Epistemic exit provides independence, institutional feedback, and protection against epistemic captivity. Re-entry allows knowledge and methods to circulate between the institution and the wider world.
Thus

$$\boxed{\text{Self-Sustainability} = \text{Internal Reproduction} + \text{External Permeability.}} \tag{326}$$

The institution must therefore be simultaneously:

$$\text{productive,} \tag{327}$$
$$\text{selective,} \tag{328}$$
$$\text{defensible,} \tag{329}$$
$$\text{criticizable,} \tag{330}$$
$$\text{permeable,} \tag{331}$$
$$\text{mobile,} \tag{332}$$
$$\text{correctable.} \tag{333}$$

Its ultimate objective is not to create an institution that never changes.
It is to create an institution that can survive precisely because it can change when evidence requires it to change.
The deepest principle of the unified SSTE is consequently:

$$\boxed{\begin{array}{c}\textbf{A self-sustaining epistemic system must reproduce its capacity without}\\ \textbf{reproducing its errors.}\end{array}} \tag{334}$$

And the corresponding institutional maxim is:

$$\boxed{\begin{array}{c}\textbf{Admit the challenger. Hear the critic. Permit the exit. Test the proposition.}\\ \textbf{Defend the method. Correct the error. Reproduce the capacity.}\end{array}} \tag{335}$$

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Glossary

Epistemologist
: An agent responsible for generating, testing, validating, preserving, reproducing, or defending epistemic capacity.

Worker
: An agent contributing informational, empirical, analytical, computational, communicative, or organizational labor to the SSTE.

Information
: A representational object entering or circulating through the epistemic process before or independently of validation.

Knowledge
: Information that has satisfied the relevant epistemic standards of its domain.

Knowledge Stock
The accumulated validated knowledge $K(t)$.

Knowledge Production
Newly validated knowledge $\Delta K(t)$ generated during a period.

Propaganda
Communicated information that has not attained validated knowledge status at the time of communication.

Epistemic Capacity
The capacity to generate, process, validate, preserve, communicate, defend, correct, and reproduce knowledge.

Epistemic Burden
The processing cost imposed by information requiring evaluation.

Epistemic Congestion
A condition in which information-processing requirements approach or exceed available epistemic capacity.

Epistemic Weapon
A methodological instrument for testing, falsifying, correcting, or neutralizing invalid inference.

Epistemic Defense
The capacity to preserve epistemic integrity and institutional function under epistemic attack.

Epistemic Attack
An action or process that degrades epistemic reliability, validation capacity, institutional memory, or decision quality.

Epistemic Reproduction
The generation of people, knowledge, methods, infrastructure, and institutions necessary to reproduce epistemic capacity.

Epistemic Depreciation
Loss of epistemic capacity through personnel loss, obsolescence, institutional deterioration, or related processes.

Epistemic Entry
Admission or temporary participation of external epistemologists or workers.

Epistemic Exit
Departure of epistemologists or workers from the SSTE.

Re-entry
Return of a former participant to the SSTE.

Epistemic Mobility
The circulation of epistemic agents through entry, participation, exit, and re-entry.

Epistemic Sovereignty
The institutional capacity to evaluate propositions through evidence, reasoning, criticism, and correction.

Epistemic Freedom
The meaningful ability to criticize an institution and to leave it.

Non-Immunity Principle
The principle that no epistemic authority, including the SSTE itself, is exempt from legitimate epistemic criticism.

Non-Sealing Principle
The principle that a self-sustaining epistemic institution must remain open to external entry, criticism, verification, and exit.

Echo Chamber
: An epistemic environment in which institutional reproduction systematically reinforces incumbent beliefs while suppressing independent variation.

Epistemic Resilience
: The capacity to absorb epistemic shocks, detect errors, recover from failures, and continue producing reliable knowledge.

Epistemic Capital
: The accumulated stock of knowledge, trained agents, methods, infrastructure, institutional memory, and defensive capability.

Epistemic Equilibrium
: A dynamic configuration in which production, validation, reproduction, depreciation, defense, entry, and exit are mutually consistent.

Viability Region
: The set of states in which the SSTE possesses sufficient knowledge, capacity, membership, and defense to remain operational.

The End